Week 7 – Design of Experiments (Lean Notes)

Lectures covered: L27–L31 (Design of Experiments, Parts 1–5, Prof. Kannan, IIT Madras): basic statistics, hypothesis testing, ANOVA, factorial designs, regression, central composite design and response surface methodology. ⭐ = tested in the Week 7 assignment.

1. Concept cards (in the order taught)

1. Why experiments, and why design them ⭐

- Experiments are done to **prove a theory**, to **get trends when theory is missing or too hard**, and as **proof of concept**. They also show the **effect of different variables on the response** (all three are correct choices in the assignment).
- Design of experiments helps you present results in a scientifically acceptable way, **screen** which variables matter (fix the unimportant ones, do fewer runs) and then find where to run next to reach the optimum (maximum yield or minimum power).
- Data first: histogram, box chart, scatter plot, normal probability plot; average the repeats; **quantify the scatter**.
- Books named: Montgomery and Runger (start here), Montgomery, Myers et al. (advanced), Ogunnaike.

2. Scatter is unavoidable: $y_i = \eta_i + \varepsilon_i$

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Terms: y_i = response in run i; η_i = true (unknown) value; ε_i = random error, can be positive or negative.

What it is for: repeats of the same experiment differ (100, 101, 97) because of the random error, so you must draw conclusions while living with scatter.

3. Random variable and probability density

- **Random variable** X (capital); its value after the experiment is x (small). A distribution predicts the behaviour of a **group** better than of one individual (a teacher can predict the class, not one student).
- **Probability density function f(x)** describes **how probability is distributed over a continuous random variable** ⭐. Total area = 1. The probability of an exact value is zero; the area between two values is the probability.

4. Cumulative distribution, mean, variance

$$F(z) = P(X \le z) = \int_{-\infty}^{z} f(x)\,dx \qquad P(a \le X \le b) = F(b) - F(a)$$

$$\mu = E(X) = \int_{-\infty}^{\infty} x\,f(x)\,dx \qquad \sigma^2 = E\left[(X-\mu)^2\right]$$

Terms: F = cumulative probability up to z; μ = mean (the centre, can be negative); σ^2 = variance (spread, always positive); σ = standard deviation (square root of variance).

What it is for: the mean says where the distribution sits, the variance how wide it is.

5. Normal distribution and standardising

- **N(μ, σ²)**: symmetric, bell shaped, mean = median = mode; μ and σ are themselves its parameters. About **68% of the area within ±1σ, 95% within ±2σ, 99% (really 99.7%) within ±3σ**. Other distributions tend to it (t when degrees of freedom → ∞).

$$Z = \frac{X - \mu}{\sigma}$$

Terms: Z = standard normal variable with mean 0 and variance 1.

What it is for: one table serves every normal distribution.

6. Population and sample

- **Population** = all entities; you cannot measure all, so take a **sample**. A good random sample has observations that are **independent**, have **equal chance** of being picked and come from the **same distribution**; a **bigger sample** gives more precise estimates (balls in a box: if the blue ones are on top, you wrongly conclude all are blue).
- A function of the sample values is a **statistic** (also a random variable).

7. Sample mean, sample variance, degrees of freedom ⭐

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

Terms: $\bar{x}$ = sample mean; s^2 = sample variance; n = sample size; n − 1 = **degrees of freedom** (the deviations add to zero, so only n − 1 are independent).

What it is for: $\bar{x}$ and s^2 are the **estimators** of μ and σ^2. Their values from one sample are **point estimates** (single numbers). - Tiny number: data 1, −1, −1, 1, 1, −1 → sum 0, n = 6, **mean 0**. ⭐ - **Units:** the mean, median, mode and standard deviation are in the data's units (cm), the variance is in cm^2. ⭐

8. Sampling distribution of the mean

$$E(\bar{x}) = \mu \qquad V(\bar{x}) = \frac{\sigma^2}{n}$$

Terms: the mean of the sample means is still μ; their variance is σ^2 divided by n.

What it is for: a bigger n means the sample means cluster tightly. If the population is normal, $\bar{x}$ is normal $N(\mu, \sigma^2/n)$ for any n. - Tiny number: $\sigma^2 = 100$, n = 25 gives $V(\bar{x}) = 4$, so the spread of $\bar{x}$ is 2.

9. Interval estimate and confidence interval

- A **point estimate** is one number (train arrives at 2:10); an **interval estimate** is a range (2:00 to 2:30).

$$\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

Terms: α = level of significance (usually 0.05); z of α/2 leaves area α/2 in each tail (1.96 for 95%); σ known; n = sample size.

What it is for: a 100(1 − α)% interval; if you took 100 samples, about 95 of the intervals would **bracket μ**. A wide interval is surer but vague; a narrow one is precise but less certain. - Tiny number: $\bar{x} = 50$, $\sigma = 10$, n = 25 gives $50 \pm 1.96 \times 2$ = **(46.08, 53.92)**.

10. Central limit theorem, t, chi-square

- **Central Limit Theorem:** even if the population is not normal, the sampling distribution of the mean tends to normal with mean μ and variance σ^2/n when n is large (say above 30).
- **t-distribution:** use when σ is **unknown and the sample is small**: $t = (\bar{x} - \mu)/(s/\sqrt{n})$, with n − 1 degrees of freedom (lecturer says "√10" by slip). Used in hypothesis tests, regression and DOE.
- **Chi-square:** $\chi^2 = (n-1)s^2/\sigma^2$ with n − 1 degrees of freedom; only positive values; used for a **confidence interval on the variance**. The tail area beyond $\chi^2(\alpha, k)$ is α (one-sided).

11. Hypothesis testing ⭐

- A claim about a **population parameter**, tested with sample data. **H_0 (null)** is the status quo (new process gives no improvement; person is not guilty); **H_1 (alternate)** is its negation. Rejecting H_0 means accepting H_1. Choose a **test statistic** (a function of the sample, such as mean or variance).
- Say "**do not reject** H_0", not "accept".

Decision	H_0 true	H_0 false
Do not reject H_0	correct	**Type II error** (less serious)
Reject H_0	**Type I error** (more serious)	correct

- Why Type I is worse: convicting an innocent person; scrapping a good process for a new one that gives no gain. A Type II error leaves you where you were.
- **p value:** the lecturer says it relates to the probability of a Type I error; a very high F gives a very small p. In DOE you start by assuming no factor matters and the data show which ones do.

12. F-distribution

$$F = \frac{s_1^2/\sigma_1^2}{s_2^2/\sigma_2^2}$$

Terms: ratio of two independent chi-squares each divided by its degrees of freedom; it has **two** degrees of freedom (numerator, then denominator); needs two normal populations.

What it is for: **comparing two variances**: is the variability from changing a factor much bigger than the variability from random noise? Example: F(0.05; 10, 10) = 2.98 (the lecturer says "2.05" by slip for α = 0.05).

13. One factor: treatments and ANOVA

- **Treatment** = a level of a factor (fertiliser A, fertiliser B). **Repeats** at the same setting give the experimental (random) error. Compare **variation between treatments** with **variation within treatments**.
- **ANOVA (analysis of variance):** sums of squares (squared so positive and negative deviations do not cancel), divided by their degrees of freedom to get **mean squares**:

Source	Degrees of freedom	Mean square	F
Treatments (a levels)	a − 1	SS treatments ÷ (a − 1)	MS treatments ÷ MS error
Error (n repeats each)	a(n − 1)	SS error ÷ a(n − 1)	

- Tiny number: a = 3 levels, n = 3 repeats gives 2 and 6 degrees of freedom. A large F (treatment mean square dominates error) means the factor matters; F near 1 means it does not.

14. Randomisation ⭐

- Do not run the settings in neat order. Photocopier example: toners A, B, C in blocks while the machine heats up would wrongly make C look bad. **Mix the order of runs** (1, 2, 3, 2, 3, 1, 1, 3, 2…) and allocate material at random so **unwanted effects spread evenly across all settings**. This is how a **true experiment controls outside variables: by randomisation**.

15. Factorial design (2ⁿ), one-factor-at-a-time

- **2ⁿ factorial**: n factors, each at 2 levels (high and low). 3 factors give 2^3 = **8 runs**. Advantages: scientific interpretation, enables response surface methods, handles **qualitative** (catalyst type) and **quantitative** (temperature) factors together, gives maximum information from limited runs; said to be essential for industrial competitiveness.
- **One factor at a time is poor:** more runs and it **misses interactions**.

16. Interaction effect (cricket example)

- Factors: bat (light or heavy) and drink (beer or tea); response = runs. Two factors **interact** when the effect of changing one depends on the level of the other. Beer + heavy bat gave the highest average (51); tea + heavy bat 24.
- **Interaction plot:** parallel lines mean **no interaction**; lines that cross, converge or diverge mean interaction. Do not rely on main-effect plots alone.
- Two-factor ANOVA adds a row for A × B with **(a − 1)(b − 1)** degrees of freedom. The error comes from repeats (all runs, or the centre point).

17. Empirical regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

Terms: β_0 = intercept; β_1, β_2 = partial regression coefficients; x_1, x_2 = factors; ε = random error with mean 0 and constant variance.

What it is for: a simple predictive formula valid **only within the range of the factors you tested**; do not extrapolate. True β are unknown; the estimates (with hats) change from experiment to experiment, which is expected. - With k factors there are **k + 1 parameters** (call it p). You need **n > k + 1 runs**; with n = k + 1 the fit is perfect but meaningless.

18. Matrix form and least squares

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon \qquad \hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

Terms: y = column of n responses; X = n × (k + 1) matrix whose first column is all ones (intercept) and other columns are the factor settings; X′ = transpose; hat = estimate.

What it is for: gives all coefficients in one step (MATLAB or Scilab). Ŷ (hat) is the predicted response.

19. Sums of squares, R^2, adjusted R^2 ⭐

$$SS_T = SS_R + SS_E \qquad R^2 = \frac{SS_R}{SS_T} = 1 - \frac{SS_E}{SS_T} \qquad R^2_{adj} = 1 - \frac{SS_E/(n-p)}{SS_T/(n-1)}$$

Terms: SST = total, SSR = regression (explained), SSE = error (unexplained) sum of squares; p = k + 1 parameters. Degrees of freedom: regression k, error n − k − 1, total n − 1. F = (SSR ÷ k) ÷ (SSE ÷ (n − k − 1)).

What it is for: R^2 (coefficient of determination) is the **fraction of variability explained**. **R^2 = 1 when SSE = 0, which is the same as SSR = SST.** (SST = 0 gives an undefined ratio; SSR = 0 gives R^2 = 0.) - R^2 = 1 is easy to cheat: use as many parameters as runs, or add useless terms (x_1^2, x_2^2). Adjusted R^2 **penalises extra parameters**; if it falls when you add a term, drop the term. A model with R^2 = 0.84 and few terms can be better than R^2 = 1 with 15 terms. - Tiny number: SST = 100, SSE = 10, n = 12, p = 3 gives R^2 = 0.90 and adjusted R^2 = 1 − (10/9)/(100/11) = **0.878**.

20. Second-order model

$$y = \beta_0 + \sum_i \beta_i x_i + \sum_{i<j} \beta_{ij} x_i x_j + \sum_i \beta_{ii} x_i^2 + \varepsilon$$

Terms: β_i = linear, β_{ij} = interaction, β_{ii} = quadratic (pure curvature) coefficients.

What it is for: a flat plane cannot show **peaks and valleys**, so optimisation needs second order terms. Two factors give 6 parameters.

21. Central composite design (CCD)

- Two factors: the **2^2 factorial square (4 runs)** + **2k axial (star) points (4 runs)** + **centre points** (repeats at the geometric centre, say 4–5). Total, for example, 4 + 4 + 5 = 13.
- **Centre points:** give cheap repeats (error), and **detect curvature**, but only tell whether $\beta_{11} + \beta_{22}$ is significant overall. **Axial points** let you estimate each **pure quadratic term** individually; they do **not** estimate interactions.

22. Response surface methodology (RSM) ⭐

- Industry cannot wander in n-dimensional space. Do **screening experiments**, then RSM points the **direction of fastest improvement** (**method of steepest ascent**), and later locates the optimum (maximum or minimum of yield, conversion, time, energy).
- Second order model in matrix form, with b the vector of linear coefficients and **B** the matrix of squares (diagonal) and cross-products (off-diagonal, halved):

$$\hat{y} = \hat{\beta}_0 + \mathbf{x}'\mathbf{b} + \mathbf{x}'\mathbf{B}\mathbf{x} \qquad \mathbf{x}_s = -\tfrac{1}{2}\mathbf{B}^{-1}\mathbf{b}$$

Terms: xs = stationary point (set the derivative b + 2Bx to zero).

What it is for: **canonical analysis**. The **eigenvalues of B** tell the type: **all negative = maximum**, all positive = minimum, mixed signs = saddle point (lecturer skips this). - Tiny number: $\hat{y} = 10 + 4x_1 + 2x_2 - 2x_1^2 - x_2^2$. Then b = (4, 2), B = diag(−2, −1), xs = (1, 1), $\hat{y} = 10 + \frac{1}{2}(4 + 2) =$ **13**; eigenvalues −2, −1 (both negative) so it is a **maximum**.

2. Top traps (most likely first)

1. **RSM canonical analysis:** all eigenvalues of B negative = **maximum**; all positive = minimum; mixed = saddle.
2. **$R^2 = 1$** when **SSE = 0** or **SSR = SST** (not when SST or SSR is zero).
3. **True experiment** controls extraneous variables **by randomisation**; incomplete designs (single group before-after) lack a control group.
4. **pdf** describes the **distribution of probability over a continuous variable** (not a scale).
5. **Mean, median, mode and standard deviation share the data's units**; variance is units squared. Mean of ±1 balanced data = 0.
6. **Type I error** (rejecting a true H_0) is the **more serious** one; say "do not reject", not "accept".
7. Variance of the sample mean is **σ^2/n**, not σ^2; the mean stays μ.
8. **Confidence interval:** 95% means about 95 of 100 such intervals bracket μ; wide = surer but vague.
9. **Use t** when σ is unknown and n small; **CLT** gives near-normal means for n above 30.
10. **n − 1** degrees of freedom for s^2; regression error has n − k − 1; interaction has (a − 1)(b − 1).
11. **Do not chase $R^2 = 1$;** check adjusted R^2; regressions hold only within the tested range.
12. **Interaction:** parallel lines = none; crossing lines = interaction; one-factor-at-a-time misses it.
13. **Centre points** detect curvature, **axial points** estimate pure quadratic terms individually and not interactions.

14. Lecturer slips: t uses √n (he says √10); α = 0.05 (he says 2.05); 99% for ±3σ is really 99.7%.

3. Assignment questions with solutions

Q1. In the true experimental research design: *(Q1)* - A. No control on external variables B. Less control on external variables C. Control on external variables by randomization D. No use of randomization

Solution (card 14): Random allocation of runs and subjects spreads outside effects evenly, which is control by randomisation. **Answer: C.**

Q2. Which of the following is an incomplete experimental research design? *(Q2)* - A. Before and after with control group design B. Factorial design C. One group before-after design D. Latin Square design

Solution (outside the lectures; table below): With a single group measured before and after, nothing separates the treatment from other influences (no control group, no randomisation). **Answer: C.**

Q3. Coefficient of determination is unity when (select all that apply) *(Q3)* - A. Error sum of squares is zero B. Total sum of squares is zero C. Regression sum of squares is zero D. Regression sum of squares is the same as total sum of squares

Solution (card 19): R^2 = SSR ÷ SST = 1 − SSE ÷ SST, which equals 1 when SSE = 0 or SSR = SST. **Answer: A and D.**

Q4. The probability density function describes the following: *(Q4)* - A. Probability of a phenomenon occurring in macro scale B. ... in micro scale C. Distribution of probabilities in the continuous random variable domain D. Distribution of probabilities in some particular set of undefined variables

Solution (card 3): Stated in the lecture. **Answer: C.**

Q5. In RSM canonical analysis with two factors, maximum response is achieved when the eigenvalues of the B matrix are *(Q5)* - A. all positive B. all negative C. all zero D. one positive and the other negative

Solution (card 22): All negative = maximum; all positive = minimum; mixed = saddle. **Answer: B.**

Q6. Experiments are conducted in laboratories to *(Q6)* - A. Verify scientific predictions B. Obtain trends when theory is not known C. Find effect of different variables on the desired response D. All of the above

Solution (card 1): The lecture gives all three reasons. **Answer: D.**

Q7. SPC and Design of Experiments are very closely interrelated. *(Q7)* - A. True B. False

Solution (outside the lectures): Statistical process control monitors a process and DOE finds the best settings, so each feeds the other. **Answer: A (True).**

Q8. Which of these is not an application of experimental design in engineering design? *(Q8)* - A. Evaluation and comparison of basic design configurations B. Evaluation of material alternatives C. Evaluation of key product design parameters that impact performance D. Evaluation of correct process to develop the product

Solution (outside the lectures): Configurations, materials and key parameters are standard uses; choosing the manufacturing process is the odd one in the answer key (remember it as the key's choice). **Answer: D.**

Q9. The average estimated from data is 50 cm. The quantity (quantities) with the same units as the data is (are) *(Q9)* - A. Standard deviation B. Median C. Mode D. All of the above

Solution (card 7): Median and mode are values from the data, and standard deviation is the square root of a squared quantity, so all are in cm (variance would be cm^2). **Answer: D.**

Q10. A binary instrument gives the sample 1, −1, −1, 1, 1, −1. The mean is *(Q10)* - A. −1 B. 0 C. 1 D. 3

Solution (card 7): Sum = 1 − 1 − 1 + 1 + 1 − 1 = 0; mean = 0 ÷ 6 = 0. **Answer: B.**

Not taught directly in the Week 7 lectures (answered from general knowledge):

Question	Fact used
Q2	a one-group before-after design is “pre-experimental” (no control, no randomisation)
Q7	SPC and DOE support each other
Q8	answer-key choice: process selection is not counted as an engineering-design application